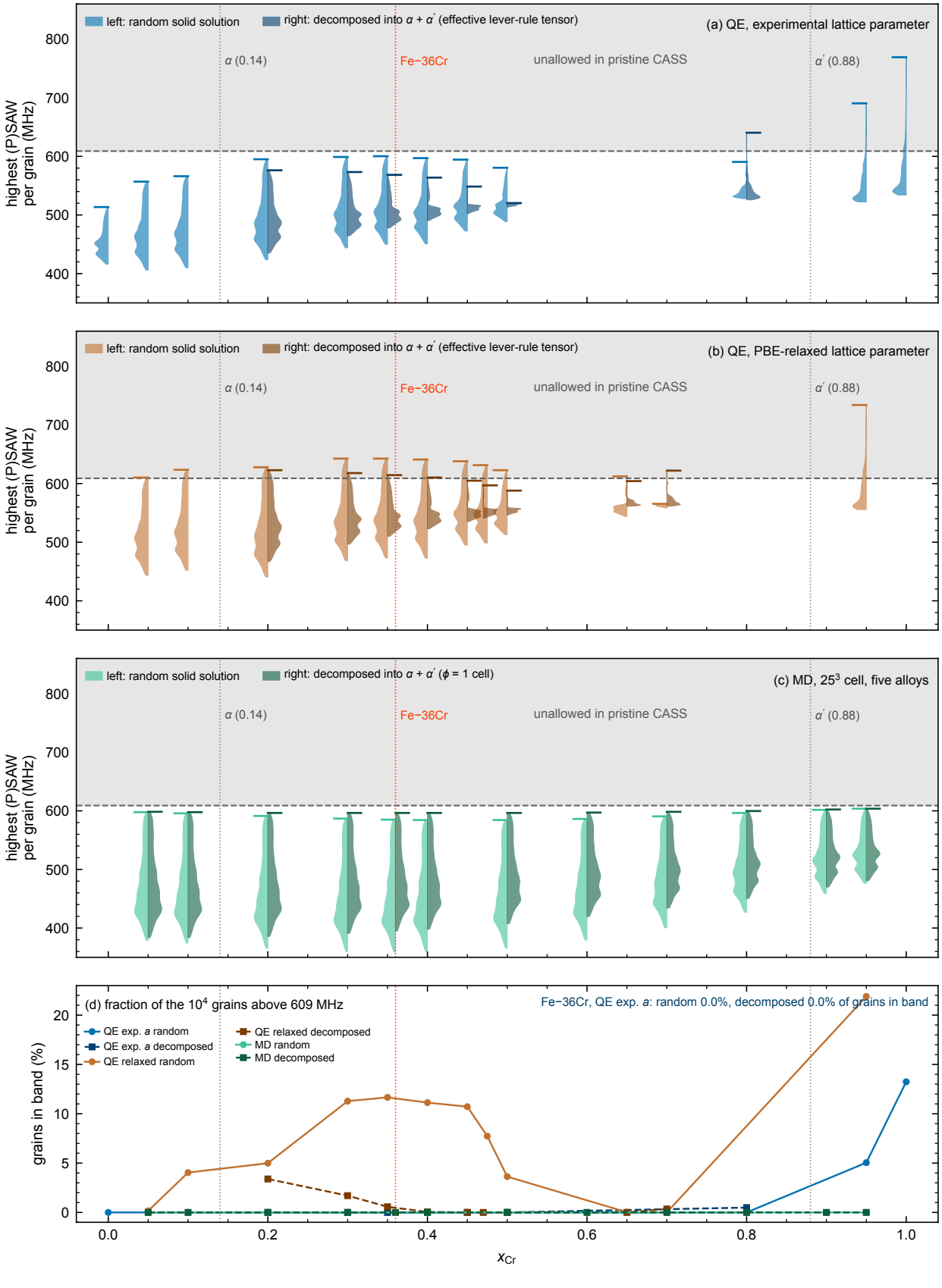

left: random solid solution
right: decomposed into α + α′ (effective lever-rule tensor)
(a) QE, experimental lattice parameter
α (0.14)
Fe−36Cr
unallowed in pristine CASS
α′ (0.88)
highest (P)SAW per grain (MHz)
800
700
600
500
400
left: random solid solution
right: decomposed into α + α′ (effective lever-rule tensor)
(b) QE, PBE-relaxed lattice parameter
α (0.14)
Fe−36Cr
unallowed in pristine CASS
α′ (0.88)
highest (P)SAW per grain (MHz)
800
700
600
500
400
left: random solid solution
right: decomposed into α + α′ (ϕ = 1 cell)
(c) MD, 25³ cell, five alloys
α (0.14)
Fe−36Cr
unallowed in pristine CASS
α′ (0.88)
highest (P)SAW per grain (MHz)
800
700
600
500
400
(d) fraction of the 10⁴ grains above 609 MHz
Fe−36Cr, QE exp. a: random 0.0%, decomposed 0.0% of grains in band
QE exp. a random
QE exp. a decomposed
QE relaxed random
QE relaxed decomposed
MD random
MD decomposed
grains in band (%)
20
15
10
5
0
0.0
0.2
0.4
0.6
0.8
1.0
x_{Cr}